

CLIENTS REQUIREMENT and IP ADDRESSNIG PLAN

CMPG 325 - COMPUTER NETWORKS

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Assigned Organization: Sebata Financial Advisory (Kimberley)

CLIENT OVERVIEW

Sebata Financial Advisory is a professional financial services firm opening a new branch in Kimberley. They handle sensitive client financial data, which requires a secure, highly available, and segregated network. The firm has requested a full network design and simulation in Cisco Packet Tracer.

BUSSNESS REQUIREMENTS

Staff operations :

The firm needs wired lab for 5 staff pcs, 1 staff laptop and 1 smartphone for daily financial operations.

Critical services :

Files, print and application servers must be available throughout business hours (08:00-17:00) without downtime.

Guest Access :

Following client change request CR3, guests in reception require isolated Wi-Fi that cannot access internal financial systems.

Secure management:

All network devices must be managed securely due to POPIA compliance.

NETWORK DESIGN -EXTENDED STAR TOPOLOGY

A central multilayer switch 3560 is deployed as the core layer to provide inter-VLAN routing (sebatacore).

- core 1x multilayer switch 3560

- Distribution 2x 2960 switches (lab switch and server room switch)

- Edge 1x router 1941 internet edge ,2x Access points 9 staff and guest)

- End devices 5x wired PCs, 1x staff laptop, 1x staff smartphone, 1x guest laptop, 1x guest smartphone ,3x servers, 1x printer.

All edge devices connect back to the core switch forming an extended star , ensuring single point of management and easy troubleshooting.

IP ADDRESSING PLAN

TOPOLOGY	VLAN ID	VLAN NAME	NETWORK	GATEWAY	SUBNET MASK	DEVICES
Topology 3	20	STAFF	172.30.62.0/25	17.2.30.62.1	255.255.255.128	PC0172.30.62.10 STAFF LAPTOP1-172.30.62.20 STAFF PHONE2-172.30.62.21
Topology 2	99	GUEST	172.30.62.128/25	17.2.30.62.129	255.255.255.128	GUEST LAPTOP-172.30.62.140 GUEST PHONE 1-172.30.62.141
Topology 1 and 2	30	SERVERS /LAB	172.30.63.0/25	17.2.30.63.1	255.255.255.128	FILE SERVER-172.30.63.10(TOPOLOGY1)

DESIGN RESTRICTION-CRITICAL SERVICE AVAILABILITY

Critical service file server is hosted in VLAN 30 (topology 1 and 4) with IP 172.30.63.10/25.

Staff VLAN 20 (172.30.62.0/250 can ping and access the server. This proves service availability despite VLAN change. Test : PC0 172.30.63.10 = success.

NETWORKING CHALLENGE -SECURE DEVICE MANAGEMENT(SSH)

Management is done via multilay switch gateway Ips. Staff VLAN 20 gateway 172.30.62.1 and guest VLAN 99 gateway 172.30.62.129 are reachable for maintenance. Test : ping 172.30.62.1 and ping 172.30.62.129 =success.

CLIENT CHANGE REQUEST -CR3 GUEST WI-FI ISOLATION

Guest network is VLAN 99 (topology 2) using 172.30.62.128/25. staff network is VLAN 20 using 172.30.62.0/25. ACL/VLAN separation prevents guest from reaching staff. This is a security requirement.

Guest laptop 172.30.62.140 ping staff PC0 172.30.62.10 = request timed out. Timeout = isolation working = CR3 Met. Guest can still ping 8.8.8.8 for internet.

TESTING AND VERIFICATION PLAN

	TEST TYPE	SOURCE	COMMAND	EXPECTED RESULT	WHAT IT PROVES
1	Intra-VLAN 20	PC0	Ping 172.30.62.20	Success4/4	Staff switch and VLAN 20 working
2	Intra-VLAN 99	GUEST Laptop	Ping 172.30.62.129	Success4/4	Guest VLAN 99 WORKING
3	Inter-VLAN	PC0(VLAN 20)	Ping 172.30.63.10	Success4/4	Routing between VLANS working
4	VLAN 30 extended	Topo1 server	Ping 172.30.63.11	Success4/4	VLAN 30 spans TOPO1 and TOPO 4
5	Security CR3	Guest .140	Ping 172.30.62.10	Request timed out	Proof: Guest isolation- guests cannot reach staff

<https://github.com/CarolMoukangwe/CMPG325-CLI091>

